

PAUSE CONTROL IN JAPANESE TEXT-TO-SPEECH CONVERSION SYSTEM WITH LEXICAL DISCOURSE GRAMMAR

Shinichi DOI, Kazuhiko IWATA, Kazunori MURAKI and Yukio MITOME

NEC Corporation, Information Technology Research Laboratories
4-1-1 Miyazaki, Miyamae-ku, Kawasaki 216, JAPAN

ABSTRACT

This paper presents a practical method for controlling prosodic information in Japanese text-to-speech conversion system based on Lexical Discourse Grammar(LDG). LDG is a method we have developed for providing a cheap and reliable parsing and global structure analyzing algorithm of Japanese long sentences with lexical information. LDG can presume the inter-clausal dependency within Japanese sentences prior to syntactic and semantic analyses by utilizing the differences of the encapsulating powers of each Japanese function words, such as conjunctive particles (postpositions) located at the end of each clause. Therefore, LDG can also produce prosodic information control parameters, especially the location and length of pauses, which are influenced by the sentence structure. We confirmed the correlation between the pause length inserted after the clause with a conjunctive particle and the particle's encapsulating power which is classified into 6 levels by analyzing the speech utterances of a male announcer.

1 INTRODUCTION

This paper introduces an approach toward controlling Japanese text-to-speech conversion based on Lexical Discourse Grammar(LDG) by presuming sentence structure with lexical information.

The length of the pause inserted in human reading is influenced by the coherence between adjoining clauses defined by the sentence structure. Conventionally, some speech synthesis methods are proposed, which control prosodic information in accordance with the sentence structure[7].

However, precise syntactic analysis of the sentence structure is hard and requires a large amount of computation, especially in Japanese. To avoid this inefficiency, we have accomplished the means to decide the location of pauses with the parts of speech of surrounding words[6]. This system is useful to output natural speech, however, it cannot handle the global structure of the input sentence, so its prosodic controlling ability is insufficient. Then prosodic information controlling method providing a module which presumes sentence structure with lexical information is desirable.

The sentence structure presumption function is also required in natural language processing system, because it can reduce syntactic ambiguities. We have developed Lexical Discourse Grammar(LDG)[8, 1], as a robust parsing technique to draw out intra-sentential discourse focusing on function words in the sentence. LDG can presume the sentence

structure prior to syntactic and semantic analyses by utilizing lexical information of function words and predicates. The Japanese function words, such as conjunctive particles and connective forms located at the end of each clause are classified by LDG into 6 levels of encapsulation powers.

In this paper, we propose a combination of the local processing and LDG to realize effective control of pause length only with surface structural processing of function words[3]. The level difference of function words is shown to have a close relationship to the pause length appearing just after the function words. To confirm the correlation between the level of a function word and the length of a pause after that word, we analyzed the speech data spoken by a male announcer. The outcome shows that the higher the level is, the longer the pause length is. We are now developing the speech synthesis system with this method.

First, conventional methods for prosodic control are overviewed. Second, LDG is explained with the characteristics for Japanese long sentences, and our proposal for pause location and length control by LDG is described. Finally, the analysis of the speech data is presented which shows the correlation between pause length and encapsulating power of function words.

2 PROSODIC CONTROL BASED ON LEXICAL INFORMATION

Speech synthesis requires the prosodic control of pause and intonation, so that it can be understood by listeners. The length of the pause inserted in human reading is known to be influenced by the coherence between adjoining clauses, that are defined by the sentence structure. Therefore, the text-to-speech conversion systems are proposed, which control prosodic information such as pause length, in accordance with the sentence structure[7].

However, even the latest technique of natural language processing cannot implement accurate analysis of sentence structures in general use. To avoid this inefficiency in sentence structure analysis, speech synthesis with the calculation of coherence between clauses by using local modification is proposed[9]. We also accomplished the means to decide the location of pauses with the information on parts of speech for surrounding words, and on nearby existence of a comma(Japanese punctuation mark *touten* “、”) [6].

Additionally, speech synthesis directly from the concept expressions can produce more reliable linguistic structures than from text. This application is also proposed for

Table 1 Conjunction Level in LDG

levels	Definition of level	Example of function words	
LEV.0	can contain every element	"to"(と=that(quotation))	higher ↑
LEV.1	cannot contain sentential particles	"kara"(から=because), "node"(ので=because), "keredomo"(けれども=but)	
LEV.2	cannot contain conclusive modal	"nara"(なら=if), "hoka"(他=besides), "to"(と=when), "baai"(場合=in case of), "toki"(時=when)	
LEV.3	cannot contain probable modal	"mae"(前=before), "ato"(あと=after)	
LEV.4	cannot contain tense expressions	"totomoni"(とともに=as), "tame"(ため=because) "todoujini"(と同時に=at the same time)	↓ lower
LEV.5	cannot contain particle "ga"(が ^s)	"nagara"(ながら=while), "tsutsu"(つつ=while), "kotonaku"(ことなく=without)	

prosodic control[4, 12]. Among them, the prosodic information calculation system[12] is effective in respect to using multiple linguistic information i.e., pragmatic, semantic, syntactic, lexical, etc., which is abstracted out of concept expressions in the description of interlingua.

3 JAPANESE SENTENCE STRUCTURE PRESUMPTION WITHOUT SYNTACTIC PARSING

3.1 Long Sentences in Japanese

As there are many long sentences with two or more predicates (i.e. compound and complex sentences), it is difficult to take the proper structure of the sentences out of an enormous number of possible modification structures. This difficulty is more marked as there are more syntactically ambiguous structures in Japanese than in English. One reason is that the Japanese language doesn't have any parts of speech to indicate either beginning or end of a phrase or a clause. Another reason is that function words to signify the sentence structure appear at the end of the clause or sentence. In short, there are few surface indicators for dividing phrases or clauses in Japanese language, unlike English, which has relative pronouns and a subordinate conjunctions.

3.2 Japanese Sentence Structure Presumption

Analyzing Japanese sentences, a huge number of possible modification structures occur out of simple parsing. A conventional bottom-up parsing method can reduce ambiguity in modification by local information in the surface structure. However, this inclines toward an improper output, as the locally highest likelihood is sometimes low on the whole.

To overcome this problem, several methods to assume global structure of the long sentences have been proposed. One is a top-down parsing method by matching the input sentence and the domain-specific patterns[5]. We also advanced and enabled this method to parse irregular, or incomplete and multiplex patterns, by describing the domain-dependent patterns in the form of grammar[2].

Another method employs global structure presumption to divide a sentence into clauses by utilizing general lexical information. It presumes the sentence structure prior to syntactic analysis by utilizing the domain-independent lexical

information such as conjunctive particles, parallel expressions, transition of the theme, etc.[11, 10]. Our LDG is one of the approaches to presume a global structure of a long sentence by focusing on function words.

3.3 Lexical Discourse Grammar(LDG)

LDG presumes the sentence structure before syntactic and semantic analyses on the basis of previously collected lexical information that characterizes the lexical discourse[8].

In Japanese complex or compound sentences, there are several dependency levels for the subordinate clause from the main one. Conjunctive particles to link clauses are classified according to the elements that the clause can contain, or to the correlation between clauses. See the following examples (* is added to meaningless sentences).

- 1) 彼は彼女が助けたので成功した。
Kare-wa kanojo-ga tasuke-ta-node
seikou-shita.
He succeeded because she helped him.
- 2)* 彼は話しながら帰った。
Kare-wa hanashi-ta-nagara kaet-ta.
He returned while he was talking.
- 3)* 彼は彼女が話しながら帰った。
Kare-wa kanojo-ga hanashi-nagara kaet-ta.
He returned while she was talking.
- 4) 彼が笑いながら尋ねたので私は答えた。
Kare-ga warai-nagara tazune-ta-node
watasi-wa kotae-ta.
I answered as he asked while he was smiling.
- 5) 彼が尋ねたので笑いながら私は答えた。
Kare-ga tazune-ta-node warai-nagara
watasi-wa kotae-ta.
I answered while I was smiling as he asked.

The clause with the conjunctive particle to express reason "node"(ので) can contain the auxiliary verb of past tense "ta"(た) and the nominative noun phrase, while the clause with the particle indicating attendant action "nagara"(ながら) cannot, as shown in 1)-3). 4) being compared to 5), the clause with "node"(ので) can subordinate the clause with "nagara"(ながら), but the reverse is impossible. Consequently, "nagara"(ながら) is ranked at a lower level than "node"(ので).

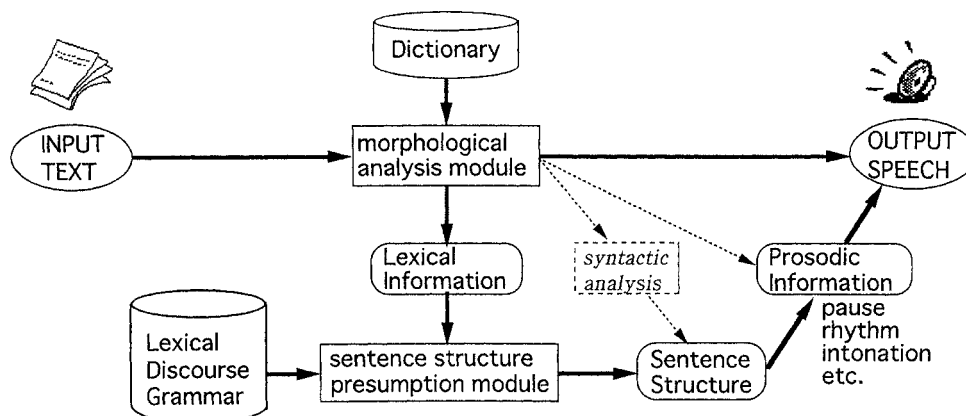


Figure 1 Prosodic Information Control in Text-to-speech Conversion System with LDG

In LDG, conjunction levels of clauses are classified into 6 by the elements the clause can contain, as listed in Table 1. These levels construct a hierarchy meaning that the clause of lower level cannot subordinate the clauses of higher levels. The levels also represent the encapsulating powers of each Japanese function words located at the end of the clauses. Besides conjunctive particles, Japanese conjunction nouns or relative nouns are also classified and assigned a level. LDG is effective to reduce the syntactic ambiguities, and we have already applied this method to our machine translation system PIVOT.

4 PAUSE CONTROL WITH LDG

As stated in the previous section, prosodic information can be accurately calculated, if the correct sentence structure is acquired for each input sentence. As yet, the newest system has not achieved the analysis in the deep structure, therefore sentence structure presumption in the surface structure is indispensable for a robust prosodic control system. LDG absolutely meets this requirement as it presumes the sentence structure by means of function words occurring on the surface. Hereafter, we propose the prosodic control system based on LDG.

As illustrated in Fig. 1, the presumption function for sentence structure (lexical discourse) by LDG is applied to pre-processing ahead of speech synthesis. It can presume the global sentence structure through lexical information without any analysis in the deep structure. It is also possible to consider the pause length inserted after each clause in relation to the lexical information in LDG. In other words, pauses are more frequently inserted after the clause of the higher conjunction levels than those of the lower levels. Consequently, the pause length and location can be more efficiently controlled with the LDG conjunction levels.

To develop the text-to-speech conversion system with LDG, it is necessary to prepare the LDG conjunction level information of a large number of conjunction equivalents such as conjunctive particles. And next, statistical data should be collected from human speech and reading, in regard to the correlations between pause length and the LDG conjunction levels. This substantial data is added to the

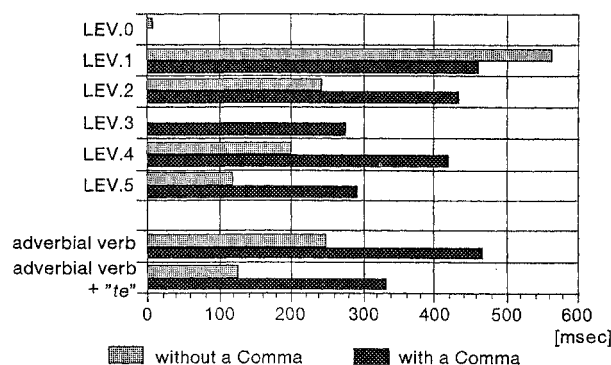


Figure 2 Average Pause Length Diagram

lexical information to be used for speech synthesis in cooperation with the pronunciation and the accent.

5 DATA ANALYSIS

To confirm the correlation between the conjunction level and the pause length, we have analyzed speech data spoken by a male announcer, reading newspapers and magazines at a regular speed. We extracted conjunctive particles and verbs, auxiliary verbs and adjectives in adverbial form from the speech data, and assorted these words by the LDG conjunction level. The average pause length for each level was calculated in two cases separately; the words preceding a comma or the words without a comma. See Figure 2 and Table 2.

This outcome shows that the higher the conjunction level is, the longer the average pause length (except for LEV.3, which is too infrequent to be estimated). This tendency does not differ, depending on the existence of a comma. We can conclude that there is a solid correlation between the LDG conjunction level and the pause length.

LEV.0 ("to"(と) and "tte"(って) functioning as quotation), however, requires a careful observation. This conjunction level, the highest rank, can contain every element

Table 2 Pause length data

Conjunction Levels	With a comma		Without a comma	
	Number of cases (with/without a pause)	Average pause length [msec]	Number of cases (with/without a pause)	Average pause length [msec]
LEV.0("to", "tte", etc.)	0	-	7 (1/ 6)	7.9
LEV.1("ga", "kara", "node", etc.)	11 (11/ 0)	461.6	3 (3/ 0)	563.3
LEV.2("ba", "to", "tara", etc.)	11 (11/ 0)	437.0	13 (11/ 2)	243.8
LEV.3("ato", etc.)	2 (2/ 0)	277.5	1 (0/ 1)	0.0
LEV.4("tame", "temo", "hodo", etc.)	5 (5/ 0)	421.5	8 (6/ 2)	201.6
LEV.5("tsutsu", "nagara", "zuni", etc.)	4 (4/ 0)	293.8	5 (2/ 3)	120.0
Verbs in adverbial form	40 (40/ 0)	468.8	6 (6/ 0)	252.1
Verbs in adverbial form + "te"(て)	14 (12/ 2)	331.8	30 (15/15)	127.4
Adjectives in adverbial form	3 (3/ 0)	542.5	19 (8/11)	89.2
Adjectives in adverbial form + "te"	1 (1/ 0)	410.0	6 (4/ 2)	56.7
Predicative auxiliary verb "da"(だ)	4 (4/ 0)	603.1	3 (3/ 0)	359.2

even an independent sentence. In this case, the relation between the conjunctive particle and its preceding clause is so weak, that pause tends to be inserted before the conjunctive particle, not after it. Therefore in the present data, a pause was inserted after the particle in only one case out of seven.

In Table 1, we didn't assign a level to two of the most frequent groups: verbs and auxiliary verbs in adverbial form, and verbs and auxiliary verbs in the same form with a conjunctive particle "te"(て). These groups are difficult to allocate to a single proper level, as they are expressing many factors such as parataxis, cause, means, attendant circumstances, and variable semantically and syntactically. However, in reference to the pause length data, the former group adverbial verbs might fall into LEV.1 or LEV.2, while the latter group with "te"(て) might fall into LEV.4 or LEV.5. Conventionally, these two groups are often treated as one "adverbial form", although many functional differences have been pointed out between them. Our data supports the difference with respect to the pause length. There are same tendencies among two adverbial forms of adjectives: ("ku"(く) and "kute"(くて)) and two adverbial forms of pseudo adjectives: ("-ni"(に) and "-de"(で)).

6 CONCLUDING REMARKS

We proposed a prosodic information control method in Japanese text-to-speech conversion system based on LDG. This method controls pause location and length in speech synthesis with conjunction level in LDG. It uses only lexical information, not requiring syntactic analysis. But it can tune the pause length finer than the method without sentence structure presumption. Analyzing speech data, we confirmed a correlation between the level of a function word and the length of a pause inserted after that word.

We are now developing the speech synthesis system with this method, through defining the default pause length to each conjunction level. Furthermore, LDG will be extended to other prosodic information (rhythm and intonation).

To refine our method, we should analyze more speech data from the various points of view. One point is relative analysis of the pause length. Because the LDG conjunction levels are closely related with the inter-clausal dependency,

LDG must be more effective when two or more conjunction equivalents of different levels appear in one sentence. As there were few such cases in the data used in this paper, we will intentionally collect such data for the proof. We should also analyze the influence of the sentence length on the pause position and length.

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